

move away from the average. Outliers are increasingly unlikely. You can safely ignore them.

This property also generates the supreme law of Mediocristan: given the paucity of large deviations, their contribution to the total will be vanishingly small.

In the height example earlier in this chapter, I used units of deviations of ten centimeters, showing how the incidence declined as the height increased. These were one sigma deviations; the height table also provides an example of the operation of “scaling to a sigma” by using the sigma as a unit of measurement.

Those Comforting Assumptions

Note the central assumptions we made in the coin-flip game that led to the proto-Gaussian, or mild randomness.

First central assumption: the flips are independent of one another. The coin has no memory. The fact that you got heads or tails on the previous flip does not change the odds of your getting heads or tails on the next one. You do not become a “better” coin flipper over time. If you introduce memory, or skills in flipping, the entire Gaussian business becomes shaky.

Recall our discussions in [Chapter 14](#) on preferential attachment and cumulative advantage. Both theories assert that winning today makes you more likely to win in the future. Therefore, probabilities are dependent on history, and the first central assumption leading to the Gaussian bell curve fails in reality. In games, of course, past winnings are not supposed to translate into an increased probability of future gains—but not so in real life, which is why I worry about teaching probability from games. But when winning leads to more winning, you are far more likely to see forty wins in a row than with a proto-Gaussian.

Second central assumption: no “wild” jump. The step size in the building block of the basic random walk is always known, namely one step. There is no uncertainty as to the size of the step. We did not encounter situations in which the move varied wildly.

Remember that if either of these two central assumptions is not met, your moves (or coin tosses) will not cumulatively lead to the bell curve. Depending on what happens, they can lead to the wild Mandelbrotian-style scale-invariant randomness.

“The Ubiquity of the Gaussian”